

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

EDMS 20-503 - 02

**SPECIFICATION
FOR
AUTOMATIC 3-PHASE METER
TESTING EQUIPMENT**

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EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

CONTENTS

CLAUSE	DESCRIPTION	PAGE
1	SCOPE	3
2	ENVIRONMENTAL CONDITIONS	3
3	CONSTRUCTIONAL REQUIREMENTS OF MTU	3
4	STANDARDS	4
5	REQUIREMENTS OF AUTOMATIC 3-PHASE METER TESTING EQUIPMENT	5
6	TEST CIRCUITS	
7	SUBMITTALS	13
8	TESTING AND INSPECTION	14
9	MARKING	18
10	Guarantee Table: MTE 3-PH. METER	20

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

1. SCOPE: This specification applies to stationary meter test units (MTUs) permanently installed in EDC laboratories' used for testing and calibration of electricity meters, in particular for their type test, acceptance test and verification test. It covers the requirements for automatic MTUs for indoor laboratory application and applies to newly manufactured MTUs to test electricity meters on 50 Hz networks with an AC voltage up to 600 V (phase to neutral).

2. ENVIRONMENTAL CONDITIONS

- The electrical and mechanical properties of the required Automatic 3-Phase Meter Testing Equipment should be guaranteed under the environmental conditions given in Table (1).

Table (1)

Ambient temperature	Maximum ambient temperature: 45 °C-(50 °C as option according EDC requirements) Minimum ambient temperature: -5 °C Maximum daily average temperature: 35 °C Maximum annual average temperature: 30 °C
Maximum humidity	relative 95 %
Altitude	Up to 1000 m above sea-level

1. General:

Constructional requirements of the MTU

- The MTU shall be of protective class I according to IEC 61140:2016. All parts which are subject to corrosion under normal working condition shall be protected effectively. Any protective coating shall not be liable to damage by ordinary handling nor to damage due to exposure to air under normal working conditions. The clearance and creepage

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

distances shall be according to IEC 60664-1:2020, overvoltage category II, pollution degree 1.

- The MTU shall be equipped with a mains switch, a mains fuse, and an emergency button.
- Single phase meter testing equipment (MTE) is required for testing and calibration of single electromechanical, static meters, prepaid meters at a time, meter types of different accuracies.
- It shall be satisfied requirements of **IEC 62057-1-2023**.

2. STANDARDS

IDENTIFIER	DESCRIPTION
IEC62057 -1-2023	Electrical energy meters – Test equipment, techniques and procedures – Part 1: Stationary meter test units (MTUs).
IEC62052-11:2003	Electricity metering equipment (A.C.) – General requirements, tests and test conditions – Part 11: Metering equipment
IEC 60417	Graphical symbols for use on equipment (available at http://www.graphicalsymbols.info/equipment)
IEC 60617	Graphical symbols for diagrams (available at http://std.iec.ch/iec60617)
IEC60664-1:2020,	IEC 60664-1:2020, Insulation coordination for equipment within low-voltage systems – Part 1: Principles, requirements and tests
IEC60721-3-2:2018,	Classification of environmental conditions – Part 3-2: Classification of groups of environmental parameters and their severities – Transportation and handling
IEC60721-3-3:2019	IEC 60721-3-3:2019, Classification of environmental conditions – Part 3-3: Classification of groups of environmental parameters and their severities – Stationary use at weather protected locations
IEC61010-031:2015,	Safety requirements for electrical equipment for measurement, control and laboratory use – Part 031: Safety requirements for hand-held probe assemblies for electrical measurement and test IEC 61010-031:2015/AMD1:2018

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

IEC61010-1:2010	Safety requirements for electrical equipment for measurement, control, and laboratory use – Part 1: General requirements IEC 61010-1:2010/AMD1:2016
IEC61140:2016	Protection against electric shock – Common aspects for installation and equipment
IEC613261:2020	Electrical equipment for measurement, control and laboratory use – EMC requirements – Part 1: General requirements
BS EN50470-1	Electricity metering equipment (A.C.) – General requirements, tests and test conditions – Part 1: Metering equipment
BSEN50470-3	Electricity metering equipment(A.C.)-Part3: Particular requirements–Static meters for active energy (class indexes A,B and C)
IEC 62055	Part 21: Framework for standardization Part 31: Particular requirements- Static payment meters for active energy (classes 1).
BS EN 61268	Alternating Current Static VAR-Hour Meters for Reactive Energy (Classes 2 and 3)
IEC62053-22:2003	Electricity metering equipment (A.C.) – Particular requirements –Part 22: Static meters for active energy (classes 0,2 S and 0,5 S)
IEC 62053-23	Electricity metering equipment (A.C) static watt hour meter for reactive energy meters class 2 & 3

3. REQUIREMENTS OF AUTOMATIC 3-PHASE METER TESTING EQUIPMENT

3.1 General:

This meter testing is required for testing and calibration of three phase electromechanical, static, or electronic meters at a time, meter types of different accuracies (class 0.2s, 0.5s, 1 and 2) can be tested as per electricity meters IEC 736, 521, 1036, 687, & 145 and the latest standard IEC 62052-11, IEC 62053-11, 21, 22, 23 ,24

- 1 phase 2 wires active power
- 3 phase 4 wires active and reactive.
- 3 phase 3 wires active and reactive.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

The test system shall be operating under circumstance:

- Mains supply voltage : $3 \times 230/400 \text{ V} + 10, -15\% \%$, $50 \text{ Hz} \pm 2 \%$
- The power supply unit uses the digital switch mode amplifiers.

3.2 Calibration test method:

* SEMI-AUTOMATIC METHOD

The operator is adjusting test console, where as the actual test procedure (calculating errors, energy control, etc ...) is automatically controlled using a reference standard meter.

I.e. error calculation by: P.C. + individual calculators or error calculation system or a microprocessor controlled meter error evaluation system as the following:

- By comparing the energy measured by the reference meter with that of the tested meter, and error calculated directly by the P.C.
- Reference meter accuracy shall be better than or equal (0.02)% for power / Energy range, and better than or equal (0.02)% for voltage/current ranges
- It shall be possible to simultaneous testing of (20) meters.
- Starting and no load testing shall be allowed.
- Testing of meters with different meter constants within the same batch shall be allowed.

3.3* AUTOMATIC METHOD

Automatic testing involved three activities which are one-time task for a type of meter. For next time operator can use the predefined table to perform automatic testing. These modules can be run stand alone or called from any other application.

a. Type definition: - In this module operator create/edit and manage the type table as per meter type. This includes the information like nominal voltage, nominal current, maximum current, meter constant and other details. This type table will be used in automatic meter testing.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

b. Editor:-Here Operator defines the test sequence which are made from small steps or group of steps to perform the specific task(s)

c. Control (Automatic test):- Here operator select the desired test sequence and run the automatic testing. After initial check programmer will inform the maximum value of test voltage and current, which will be applied during the test. operator is allowed to proceed or cancel the testing at any stage of testing. At the end of testing result can be stored, printout or display at the monitor.

It shall include at least :

a- Reference meter and error calculation system :-

The test system satisfies IEC 62052-11, IEC 62053-11, 21, 22, 23,24 accuracy of reference meter shall be (0.02) % or better for power range, and (0.02) % for voltage/current ranges; directly connected in the test circuit, without using measuring transformers or transformer incorporated in it, so that the accuracy of the reference meter shall be the same as the accuracy of the whole test system, the reference meter shall have a display to display all measured and calculated parameters.

- Reference standard voltage range: 30–400 V (Phase- neutral).
- Reference standard current range: 1 mA– 120 A (direct)
- The reference standard meter shall have a LCD display (safely connected to the reference meter) and have a mean of display reference meter, real -time reading for voltage, current, P.F and power (P,Q,S).

b- Meter Suspension rack (s) :-

For (20) meters to be tested together, including supports for sensors (20) scanning heads

Sensor shall satisfy:

- Ease of alignment of scanning Heads
- Suspension system easily adjusted for meter access and scanning head positioning.
- Insensitive to sun light.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

- Scanning head devices shall be capable to sense the rotor mark and led blinking of new electronic meters.
- The scanning head holding arrangement shall be easily.
- Adjustable vertically and horizontally.
- The scanning heads shall be suitable for testing both induction and static meters
- The wave length of the received signal from under test static meters that can be received and evaluated by the sensors shall be within the range 500-950 nm.

c- Selection and adjustment devices:

For voltage and current: The voltage and current circuits shall be adjustable and permit all load and no-load calibration acc. to IEC "62053-11" and "62053-23" also it shall has the capability of testing meters with closed link between current and voltage circuit.

d- Error indication unit:

Each measurement position shall have individual error LCD display unit. Error display unit shall be connected in such away that any number of meters (max up to 20 meters) can be tested at a time. In case one device is not working or not in use still error should be available at P.C.

- e- Error determination of the test system:** according to IEC 62052-11 and according to accuracy offered.

3.4 The test bench:

(A) : The test bench shall include the necessary constructional requirements and requisite connections designed to enable KWH meters cl. 0.2S & cl. 0.5s & cl. 1.0 & cl. 2.0 – KVARH meters cl. 0.5s & cl. 1.0 & cl. 2.0 to be tested under reference conditions specified in IEC 62053-11, 62053-23 resp.

The wiring shall be such as to minimize effects of magnetic induction, capacitive interference and voltage drop. Easy access to all elements is allowed by dismounting side, front and rear panels.

(B) : The test bench shall be possible to open the current circuit to prevent any current being produced by induction in these circuits by the voltage circuits.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

(C) : Adjusting devices:

(C1) Voltage Amplifier:

- The voltage amplifier shall permit the adjustment of the rated voltages (3 x 230/400) v for 3 phase, 3 & 4 wires active and reactive meters &
- ((3*63.5)/110)V for 3 phase, 3 & 4 wire, active and reactive meters for cl.0.2S & cl. 0.5s & cl.1.0 & cl.2.0 KW.H meters in IEC 62053-11 and cl. 0.5s & cl.1.0 & cl. 2.0 KVARH meters in IEC 62053-23. And it shall permit adjusting the necessary testing voltages with 0.05% accuracy for electronic test voltage generation.
- The voltage amplifier is electronically protected against overload as well as short circuit and has galvanic isolation between mains and output

C2. Current Amplifier:

- The current amplifier shall permit the adjusting of the necessary testing current within the entire load range for which the test system is provided and within the tolerance given for KWH & KVARH meters. And it shall permit adjusting the necessary testing current (120 A) with 0.05% accuracy for electronic test current generation.
- The current amplifier shall be electronically protected against overload as well as open circuit and has a galvanic isolating between mains.

C3. Phase angle adjustment:

The phase angle device shall permit fine adjustment of the phase angle within 0°.... 360°.

Resolution: 0.01°

Setting accuracy: 0.01°.

- Also the device shall permit quick section of $\cos \Phi = 1.0$ to $\cos \Phi = 0.5$.
- The adjustment shall be within the tolerances given for cl.0.2S & cl. 0.5s & cl.1.0 & cl.2.0 K.W.H meters and cl. 0.5s & cl.1.0 & cl.2.0 KVARH meters.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

- There shall be a built in watt meters or P.F meter for P.F checking.
- The ability of a phase sequence change over software and indication of inverse phase sequence
- Shall permit continuously adjusting the phase angle P.F between test voltage & test current within $\pm 120^\circ$ (continuously Step $\leq 0.01^\circ$).

(D) Voltage and current Amplifiers:

- Shall allow compliance with permissible limits of percentage for testing KWH & KVARH meters according to IEC 62053-11 / 62053-23-24.
- (separated or combined) voltage and current amplifiers are accepted if the max output current & voltage at all positions simultaneously with no influence between the accuracy of current in whole range of voltage & no influence between the accuracy of voltage in whole range of current & should be tested for the whole positions during factory acceptance test for 120 minutes.
- The amplifiers shall be switch mode amplifier and transformer less

(E) Measuring parameters:

(E1) The voltage:

Measuring of the rated voltage are within the tolerance given in the reference conditions for cl.0.2S & cl. 0.5s & cl.1.0& cl.2.0 KWH meters (IEC 62053-11) and cl. 0.5s & cl.1.0& cl.2.0 KVARH meters (IEC 62053-23) within 0.02% accuracy.

(E2) The current:

Measuring of the rated current are within the tolerance given in the reference conditions for cl.0.2 & cl. 0.5s & cl.1.0& cl.2.0 KWH meters and cl.0.5S & cl.1.0 & cl.2.0 KVARH meters within 0.02% accuracy.

(E3) The power:

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

Measuring and adjustment of the power between the permissible limits for cl.0.2 & cl.0.5s & cl.1.0& cl.2.0 KWH, cl.0.5s & cl.1.0& cl.2.0 KVARH meters, depending also on the required value within 0.02% accuracy.

(E4) Phase angle displacement indication or display:

- Scale graduations (if analog indication) or resolution (if digital display) shall be 0.01° , with setting accuracy 0.01° .
- **Note:** above stated measuring instruments or displays for V, I, Q, frequencies are required regardless of any displays by the computer.

(F) Transmission of Meter's Data:

The test system should be provided by optional hand held terminal to collect meter related data in meter test bench and transfer it to the control software for further processing according to EDC requirements.

1- Test circuits:

(A) Test current circuit:

- At least testing currents required for testing 3-phase KWH, KVARH meters shall be allowed for testing , and all these test current shall be clearly indicated digitally or on graduation of meter scale. Test current ranges shall be at least 9 ranges with stepless adjustment within at least 0-100% of each current range selected. The current range shall include most of the basic currents of test meters e.g. 0.1, 0.25, 0.5, 2.5, 5, 10, 25, 50, 100, 120 A.
- Power output shall be ≥ 2000 VA per phase with accuracy of 0.05% and protected against open circuits.
- Current range: 1mA to 120A.
- Accuracy of current circuit: 0.05%.
- Superposition of harmonics : 2.....21th harmonics with adjustable magnitude
- Push buttons for switching ON/OFF of test current and test voltage or for selecting of test currents and voltages shall be with indicating lights.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

- It is allowed to use the keys of the computer keyboard & software instead of push buttons and adjusting or selecting or change our switches are specified for switching ON/OFF of any circuit for any other functions allowed to any push button in this case no indicating lights are required.
- The MTE shall allow individually switching or adjusting of test current in each phase to allow calibration element by element or all elements simultaneously.

(B) Test voltage circuit:

- Step less adjustment of test voltages shall be allowed at least 80-120% for each U_n (nominal voltage) (3 x 230/400 V) for 3-phase, 4 wire active and reactive & (63.5/110 $\sqrt{3}$) V for 3-phase, 3 wire active and reactive.
- Power output shall be ≥ 500 VA per phase with accuracy of 0.05% and protected against short circuits.
- Voltage range: 30...300V (phase to neutral)
- Superposition of harmonics: 2.....21th harmonics with adjustable magnitude over
- Accuracy of voltage circuit: 0.05%.
- It is allowed to use the keys of the computer keyboard & software instead of push buttons and adjusting or selecting or change our switches are specified for switching ON/OFF of any circuit for any other functions allowed to any push button in this case no indicating lights are required.

(C) Connection of meters to be tested:

- Manual connection shall make possibility for testing more than one type of meters and the cables must be suitable to bear the flow of high current up to 120 Ampere without over heating or damage.

(D) Safety elements:

- Necessary protection against overload and short circuiting on MTE components (e.g. reference meter,etc) shall be taken care of.

(E) A provision for testing closed link single-phase meter:

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

A multi secondary voltage transformers with 1 primary winding (110V...300V) and 21 secondary winding (110V...300) which is capable of testing 20 **single** phase closed link meters at the same time and supplying isolated voltage for reference meter to measure the test voltage in the circuit for accuracy measurement, so single position voltage transformer is not accepted. (MSVT) as option according to EDC requirements

- Primary voltage 110V...300V, 50...60 Hz
- Secondary voltage 110V...300 V
- MSVT VA 15 VA

(F) A provision for testing closed link three phase meter (for N positions acc. To EDC requirements):

- The precision three-phase isolation current transformer ICT can be used to separate the generated test current. This can be used if meters with an irremovable connection between current and voltage might be tested. The three-phase ICT provides the testing for single-phase and three-phase electricity meters. The ICT should be equipped with pluggable secondary cables to insure closing the secondary circuit perfectly and prevent any open circuit and its hazards.

Technical parameters	Requirements		
Maximum current (I_{max})	120 A per phase		
Minimum current	10 mA per phase (optionally 1 mA)		
Ratio	1:1		
Power rating	Min. 60 VA per phase at 120 A, max. output voltage 0,5 V over the whole current range.		
	Current range	Maximum Ratio error %	Maximum Phase angle error, min
Accuracy	$1 \text{ mA} \leq I < 10 \text{ mA}$	1	50
	$10 \text{ mA} \leq I < 25 \text{ mA}$	0,5	20
	$25 \text{ mA} \leq I < 150 \text{ mA}$	0,2	10
	$150 \text{ mA} \leq I \leq 120 \text{ A}$	0,05	3
Power supply of the ICT for compensation purpose	Shall be decided by the manufacturer		

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

SUBMITTALS:

The following documents shall be supplied with the offer: -

- A complete instructor guide describing the mode of operation of the whole equipment.
- A detailed list of a apparatus and accessories.
- Catalogues and detailed description that shows compliance with all above requirements.
- The attached guarantees schedule duty complete and signed.
- Offers not accompanied with catalogues and schedule of guarantee as described above shall not be considered.

4. Testing and inspection:

All tests required to prove compliance with IEC 62052-11 requirements, shall be carried out at the expense of the supplier by the manufacturer.

- A- EDC has the right to send its representative(s) to attend such tests. Testing dates shall be advised by the contactor to permit witnessing by EDC representative(s). Testing certificates shall be submitted to EDC
- B- Unless otherwise accepted in writing by EDC no goods shall be shipped before inspection and testing and acceptance of EDC representative(s) for such equipment.
- C- Testing in the presence of EDC representative(s) shall not relive supplier from his full responsibility about supplying goods according to all above specifications.
- D- Testing shall also include any test EDC shall see necessary to prove correctness of guarantee table values for the MTE or any of its components, especially the VA power output of the testing current and voltage circuits (i.g. testing of voltage transformer and phase shifter by loading them by the guarantee table values of VA for a long period regardless the number of test meters specified as 20 meters also electric generators VA may be tested.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

5. **GUARANTEE:**

- Supplier (contractor) shall guarantee supplies against all defects arising out of faulty design manufacture, or defective material for three years from date of delivery to EDC stores for supplier's with previous supply to EDC .
- In case of supplying for the first time ,the guarantee period shall be extended to five years at least from delivery date.

6. **EQUIPMENT AND ACCESSORIES:**

A- The various subassemblies shall be easy access to facilitate maintenance.

B- The metallic equipment shall be treated with anti-corrosion.

- The testing equipment must be constituted of all equipment and accessories needed to perform tests mentioned above including (the test voltage cables, the test current cables set for whole current 10/60/120A with pluggable pin & cables set for CT & VT)

C- The following components of the MTE shall be manufactured by the same MTE manufacturer.

- 1- Reference standard meter.
- 2- Scanning head and error indicators.
- 3- Frequency wave form generation.
- 4- Current and voltage amplifiers (for MTE with electric test power generation) the amplifiers shall be transformers less.

D- The test bench has the capability to test other portable reference meters with different accuracies.

N.B:

The test bench and amplifiers manufacturer should be the same manufacturer of the reference meter

- The manufacturer should have a reference list for the delivery of the same quoted bench to any electrical utility or over the world and he should supply us with a copy of the purchasing order.

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

- The supplier must provide proof of technical competence to repair MTU after the end of the warranty period, such as providing maintenance orders and maintenance contracts for this type of MTU for worldwide with the possibility of field visits to sites under the maintenance of the supplying company
- The country of origin should be Europe, Japan, or USA and the manufacturer should submit a certificate of origin and arrange a training visit and an inspection visit for at least two engineers.
- The manufacturer should have an agent in Egypt with original **S14** certificate; the agent should submit the original **S14** certificate on request for matching with submitted copy.

7. TECHNICAL SPECIFICATIONS:

No. of testing meter	20
Supply	3*230/400 +10 , -15%
Voltage frequency	50 HZ $\pm$ 2%
Three phase voltage amplifier:	3*30....300 V (P-n)
- Voltage range	
- Power	$\geq$ 500 VA(per phase)
- Protection	Against short circuits
- accuracy	$\leq$ 0.05%
Three phase current amplifier:	3*1 mA...120A
- Current range	
- Power	$\geq$ 2000 VA (per phase)
- Protection	Against open circuits
- accuracy	$\leq$ 0.05%

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

Reference meter with display: - Accuracy of power - Measurement current range - Accuracy of current range - Measurement voltage range - Accuracy of voltage range	Not more than 0.02% 3*1 mA.....120A ≤ 0.02% 3*30.....400V (p-n) ≤ 0.02%
Phase angle: - Phase shifting - Resolution and accuracy - Step	From 0° to 360° 0.01° 0.01°
Harmonic generation of current and voltage amplifiers: - Harmonics	2nd to 21th
Connection cables: - Type of connection - Connection with source - Connection with PC	Free wires Yes Yes
Operating conditions: - Temperature - Relative humidity	From 0° to 50° Up to 95%

8. Test Bench Software capabilities

The software of the test bench should have the following capabilities:

- Pre-warming

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

- Accuracy testing of energy meters
- Starting Current Test
- Creep Test
- Influence quantity test as per IEC 62053-11,21, 22,23,24 (which are mainly linked with accuracy test)
- Dial test
- Register test
- Influence of self-heating
- Harmonics generations (Harmonics up to 2nd to 21th)
- Sub-Harmonics and odd & even Harmonics generation
- Influence of supply voltage: (Voltage interruption/ Voltage dips / severe voltage variation) test
- Immunity of earth fault
- No load conditions
- Meter constant test
- Absolute measurement or Testing of reference meter class 0.05

9. MARKING

- Each test equipment shall be provided with an easily readable nameplate
- Each nameplate will include at a minimum the following:
 - i.No. of positions
 - ii.manufacturer's name
 - iii.Country of Origin
 - iv.date of manufacturing
 - v. accuracy class;
 - vi. serial number;

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

- vii. month and year of manufacturing (not coded);
- viii. number of phases;
- ix. rated VA;
- x. rated current;
- xi. rated voltage
- xii. rated frequency;
- xiii. Purchase order Number (P.O. Number).

10. Training

TRAINING and CUSTOMER SUPPORT:

- The supplier shall provide adequate on-site training for a sufficient number of...EDC representatives, to be agreed by (...EDC) and supplier.
- The training should include and not limited to the following:
 - i. Installation and Programming.
 - ii. Testing and Commissioning.
 - iii. Troubleshooting.
 - iv. Operating and Maintenance
- The Supplier must have a dedicated service center and support line in Egypt to cover Test equipment's benches.
- EDC reserves the right to claim a free after-sales service for the entire warranty

Period. After-sales service should include and not limited to the following:

- i. Technical support.
- ii. Maintenance.
- iii. Original Spare parts.
- All services, spare parts, materials and accessories required during the after-sales

Maintenance should be delivered by the supplier immediately wherever required (Warehouse, site, laboratory, etc.)

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

11. Guarantee Table: MTE 3-PH. METER

1. **Type and model**.....
2. **Country of manufacture**.....
3. **Standards which testing equipment satisfies**.....
4. **Test current circuit: -**
 - ❖ Current output..... VA
 - ❖ Current range..... A
5. **Test voltage circuit:**
 - ❖ Voltage output (voltage circuit output power) VA
 - ❖ Voltage range..... V
 - ❖ Stepless adjustment of test voltage within.....% of selected voltage ranges.
 - ❖ Push button for ON/OFF or for selecting of test voltage illuminate. YES () NO ()
 - ❖ Quick selector for 80-100-120% of selected voltage range. YES () NO ()
6. **Phase adjustment:**
 - ❖ Step less adjustment of power within zero to $\pm$ degree
 - ❖ Operation of phase shifter by (hand / motor)
 - ❖ Indication of phase shift by (KW meter or phase angle meter).
 - ❖ Quick selector of $\cos \emptyset = 1.0$ to $\cos \emptyset = 0.5$ YES () NO ()
 - ❖ A phase sequence changeover switch..... YES () NO ()
 - ❖ An indication of inversed phase sequence..... YES () NO ()
7. **Reference standard meter:**
 - ❖ Type, model.....
 - ❖ Accuracy Active energy (P.F = 1).....
 - Active energy (P.F = 0.5).....
 - ❖ Accuracy Reactive energy ($\sin \emptyset = 1$).....
 - Reactive energy ($\sin \emptyset = 0.5$).....
 - ❖ Error calculation system:
 - ❖ Type, model and manufacturing company
 - ❖ Capacity (active and reactive meters)
8. **Suspension Rack:**

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 20-503 - 02
Automatic 3-Phase Meter Testing Equipment	DATE :29-11-2023

- ❖ Built in or separate
- ❖ Capacity: number of 3-phase meters to hang.....
- ❖ Height of the horizontal bar adjustment range.....
- ❖ Has a rail to carry scanning heads..... YES () NO ()
- ❖ No. of scanning heads supplied with rack.....
- ❖ No. of individual calculators at test positions.....

9. Scanning heads for testing meters of mech.type:

- ❖ Type, model.....
- ❖ Ease of alignment of photo-cell pick-up, pick-up arms easily swung out for meter access..... YES() NO()
- ❖ Scanning heads suitable for mech. and static meters..... YES() NO()
- ❖ Focal distance from sensor to meter..... mm
- ❖ Heads insensitive to sun light..... YES() NO()
- ❖ Diodes color

10. Main connection:

- Unstablized mains, 50 HZ.....Volt

11. Scanning head for testing KWH – meters of mech. type:

Type.....

Ease of alignment of photo-cell pick-up, pick-up arms easily swung out for meter access.....YES () NO ()